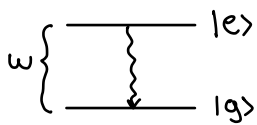


TLS con forzamiento

El hamiltoniano del sist. sin forzamiento es $H_{sf} = \underbrace{\hbar\omega |e\rangle\langle e|}_{\text{Sist. aislado}} - \underbrace{i\hbar\gamma |e\rangle\langle e|}_{\text{Decaimiento}}$



Notación: $|g\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|e\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $\overset{\text{Bajada}}{\sigma_-} = |g\rangle\langle e|$, $\overset{\text{subida}}{\sigma_+} = |e\rangle\langle g|$, $\sigma_z = \sigma_+\sigma_- - \sigma_-\sigma_+$

Introducimos el forzamiento coherente:

$$H = \hbar\omega \sigma_+\sigma_- + \hbar g (\sigma_+ + \sigma_-)$$

Energía de 1 TLS Acoplamiento coherente Decaimiento por el entorno
↑ ↑ ↑

$$K = \hbar\omega \sigma_+\sigma_- + \hbar g (\sigma_+ + \sigma_-) - i\hbar\gamma \sigma_+\sigma_-$$

$$\sigma_+ + \sigma_- = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} (1,0) + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} (0,1) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma_x$$

Ec. maestra de Lindblad:

$$\begin{aligned} \dot{\rho} &= \mathcal{L}\rho = \frac{1}{i\hbar} [H, \rho] + \gamma (2\sigma_- \rho \sigma_+ - \sigma_+ \sigma_- \rho - \rho \sigma_+ \sigma_-) \\ &= \frac{1}{i\hbar} (H\rho - \rho H) + 2\gamma \sigma_- \rho \sigma_+ - \gamma \sigma_+ \sigma_- \rho - \gamma \rho \sigma_+ \sigma_- \\ &= \frac{1}{i\hbar} H\rho - \frac{1}{i\hbar} i\hbar\gamma \sigma_+ \sigma_- \rho - \frac{1}{i\hbar} \rho H - \frac{1}{i\hbar} i\hbar\gamma \rho \sigma_+ \sigma_- + 2\gamma \sigma_- \rho \sigma_+ \\ &= \frac{1}{i\hbar} \left(\underbrace{H\rho - i\hbar\gamma \sigma_+ \sigma_- \rho}_{(H - i\hbar\gamma \sigma_+ \sigma_-)\rho} - \underbrace{\rho H - i\hbar\gamma \rho \sigma_+ \sigma_-}_{\rho(H + i\hbar\gamma \sigma_+ \sigma_-)} \right) + 2\gamma \sigma_- \rho \sigma_+ \\ &= \frac{1}{i\hbar} (K\rho - \rho K^\dagger) + 2\gamma \underbrace{\sigma_- \rho \sigma_+}_{(\sigma_- \otimes \sigma_+^\dagger)\rho} \\ \Rightarrow \mathcal{L} &= -\frac{i}{\hbar} (K \otimes \mathbb{1} - \mathbb{1} \otimes K^\dagger) + 2\gamma \sigma_- \otimes \sigma_+^\dagger \end{aligned}$$

$$\begin{aligned} K &= \hbar\omega \underbrace{\sigma_+\sigma_-}_{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}} + \hbar g \underbrace{(\sigma_+ + \sigma_-)}_{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}} - i\hbar\gamma \underbrace{\sigma_+\sigma_-}_{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}} = \begin{pmatrix} 0 & 0 \\ 0 & \hbar\omega \end{pmatrix} + \begin{pmatrix} 0 & \hbar g \\ \hbar g & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & i\hbar\gamma \end{pmatrix} \\ &= \begin{pmatrix} 0 & \hbar g \\ \hbar g & \hbar(\omega - i\gamma) \end{pmatrix} \Rightarrow K^\dagger = \begin{pmatrix} 0 & \hbar g \\ \hbar g & \hbar(\omega + i\gamma) \end{pmatrix} \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 0 & g \\ g & \omega + i\gamma \end{pmatrix} \end{aligned}$$

$$\Rightarrow K \otimes \mathbb{1} = \begin{pmatrix} 0 & 0 & \hbar g & 0 \\ 0 & 0 & 0 & \hbar g \\ \hbar g & 0 & \hbar(\omega - i\gamma) & 0 \\ 0 & \hbar g & 0 & \hbar(\omega - i\gamma) \end{pmatrix}, \quad \mathbb{1} \otimes K^\dagger = \begin{pmatrix} 0 & \hbar g & 0 & 0 \\ \hbar g & \hbar(\omega + i\gamma) & 0 & 0 \\ 0 & 0 & 0 & \hbar g \\ 0 & 0 & \hbar g & \hbar(\omega + i\gamma) \end{pmatrix}$$

$$K \otimes 1 - 1 \otimes K^\dagger = \hbar \begin{pmatrix} 0 & -g & g & 0 \\ -g & -\omega - i\tau & 0 & g \\ g & 0 & \omega - i\tau & -g \\ 0 & g & -g & -2i\tau \end{pmatrix} \Rightarrow -\frac{i}{\hbar} (K \otimes 1 - 1 \otimes K^\dagger) = -i \begin{pmatrix} 0 & -g & g & 0 \\ -g & -\omega - i\tau & 0 & g \\ g & 0 & \omega - i\tau & -g \\ 0 & g & -g & -2i\tau \end{pmatrix}$$

$$\sigma_- \otimes \sigma_+^\dagger = \sigma_- \otimes \sigma_- = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow 2\tau \sigma_- \otimes \sigma_+^\dagger = \begin{pmatrix} 0 & 0 & 0 & 2\tau \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow \mathcal{L} = -\frac{i}{\hbar} (K \otimes 1 - 1 \otimes K^\dagger) + 2\tau \sigma_- \otimes \sigma_+^\dagger = -i \begin{pmatrix} 0 & -g & g & 2i\tau \\ -g & -\omega - i\tau & 0 & g \\ g & 0 & \omega - i\tau & -g \\ 0 & g & -g & -2i\tau \end{pmatrix}$$

$$= \begin{pmatrix} 0 & ig & -ig & 2\tau \\ ig & i\omega - \tau & 0 & -ig \\ -ig & 0 & -i\omega - \tau & ig \\ 0 & -ig & ig & -2\tau \end{pmatrix}$$

$$\det(\mathcal{L} - \lambda 1) = \det \begin{pmatrix} -\lambda & ig & -ig & 2\tau \\ ig & i\omega - \tau - \lambda & 0 & -ig \\ -ig & 0 & -i\omega - \tau - \lambda & ig \\ 0 & -ig & ig & -2\tau - \lambda \end{pmatrix}$$

$$= -\lambda \{ (i\omega - \tau - \lambda) [(-i\omega - \tau - \lambda)(-2\tau - \lambda) + g] - ig [(i\omega + \tau + \lambda)(-ig)] \}$$

$$-ig \{ ig [(-i\omega - \tau - \lambda)(-2\tau - \lambda) + g] - ig [g] \}$$

$$-ig \{ ig [-ig] - (i\omega - \tau - \lambda) [(-ig)(-2\tau - \lambda)] - ig [-g^2] \}$$

$$+ 2\tau \{ ig [(i\omega + \tau + \lambda)(-ig)] + (i\omega - \tau - \lambda) [g^2] \}$$

$$= -\lambda \{ (i\omega - \tau - \lambda) [i2\tau\omega + i\omega\lambda + 2\tau^2 + \tau\lambda + 2\tau\lambda + \lambda^2 + g] - i\omega g^2 + \tau g^2 + g^2\lambda \}$$

$$+ ig^3 [i2\tau\omega + i\omega\lambda + 2\tau^2 + \tau\lambda + 2\tau\lambda + \lambda^2] \quad 3\tau\lambda$$

$$+ ig^3 [-ig - (-i2\tau\omega - i\omega\lambda + 2\tau^2 + \tau\lambda + 2\tau\lambda + \lambda^2 + g^2)] \quad 3\tau\lambda$$

$$+ 2\tau (i\omega g^2 + \tau g^2 + g^2\lambda + i\omega g^2 - \tau g^2 - g^2\lambda)$$

$$= -\lambda (-2\tau\omega^2 - \omega^2\lambda + i2\tau\omega\tau^2 + i3\omega\tau\lambda + i\omega\lambda^2 + i\omega g$$

$$- i2\tau^2\omega - i\omega\tau\lambda - 2\tau^3 - 3\tau^2\lambda - \tau\lambda^2 - \tau g$$

$$- i2\tau\omega\lambda - i\omega\lambda^2 - 2\tau^2\lambda - 3\tau\lambda^2 - \lambda^3 - g\lambda$$

$$- i\omega g^2 + \tau g^2 + g^2\lambda)$$

$$+ ig^3 (i2\tau\omega + i\omega\lambda + 2\tau^2 + 3\tau\lambda + \lambda^2 - ig + i2\tau\omega + i\omega\lambda - 2\tau^2 - 3\tau\lambda - \lambda^2 - g^2)$$

$$+ 2\tau (2i\omega g^2)$$